# Check Your Lens Before You Guard:

## Jacobian vs. Logit Lens for Jailbreak Monitoring in an SLM

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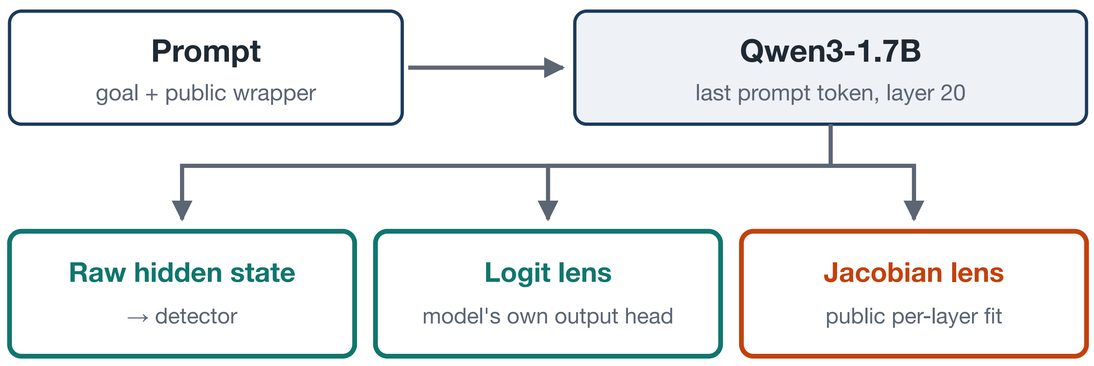
**Q1** Is a **public lens** faithful at the position where a guard reads?

**Q2** Does the **raw hidden state** there predict a jailbreak's outcome, beyond cheap text shortcuts?

### Background

A **pre-generation guard** built on a small language model (SLM) reads the model's state *after the prompt and before the first response token*, so a jailbreak can be stopped before any harmful text exists. Public interpretability lenses are an attractive, explainable readout for such a guard, but only if they are faithful where the guard reads.

### Method




- **One read** at the last prompt token, before the first response token.
- **Lens score:** top-10 agreement with the model's own output at the chat-template position.
- **Model:** Qwen3-1.7B (28 layers; float32, enable_thinking=False, greedy).
- **Data:** JailbreakBench and HarmBench goals (IDs only), two StrongREJECT wrapper families, provisional labels from Qwen3Guard. Goal-grouped 60/20/20 split; **the frozen test split was not read**.
- **Analysis:** goal-grouped 5-fold CV on train+dev (5 seeds, 140 configs), attack-only *within-family* AUROC, with text baselines and controls.
- **Pre-registered:** all choices and failure criteria were written down and time-stamped before any result was seen.

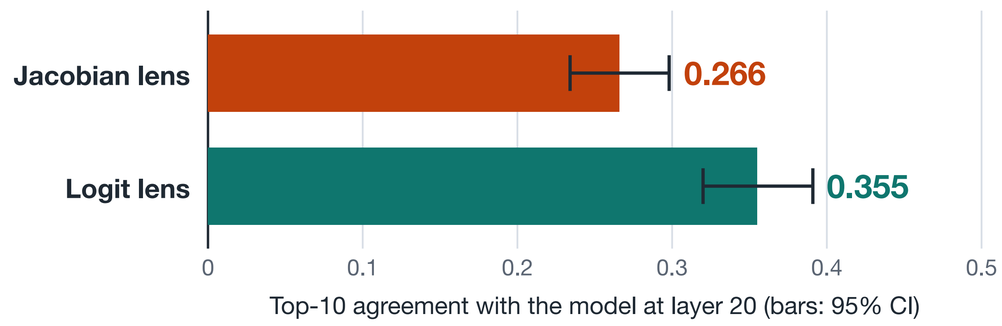
### Table I. Within-family AUROC train+dev CV, mean [min, max], 5 seeds

| Feature | Best setting | AUROC |
|---|---|---|
| **Raw hidden state** | layer 20 | **0.761** [0.729, 0.792] |
| Refusal-direction projection | layer 20 | 0.659 [0.595, 0.724] |
| TF-IDF | unigram | 0.539 [0.520, 0.580] |
| Prompt length | – | 0.398 [0.346, 0.470] |
| Template ID | – | 0.381 [0.337, 0.445] |
| Random features (control) | 20 seeds | 0.496 (p95 0.562) |
| Shuffled labels (control) | 20 seeds | 0.477 (p95 0.582) |

### Finding 1 · The raw hidden state predicts the outcome

**0.761** within-family AUROC at layer 20 (preliminary), above every text baseline (best 0.539) and both controls (Table I).

### Finding 2 · The Jacobian lens fails where the guard reads



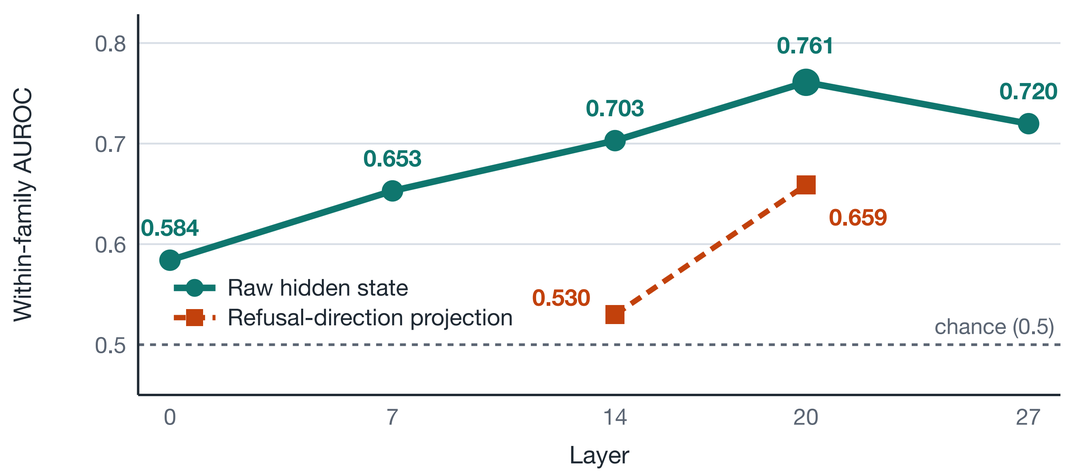

Top-10 agreement with the model at layer 20 (bars: 95% CI)

At layer 20 the public Jacobian lens agreed with the model **less** than the plain logit lens: **0.266** [0.234, 0.298] vs. **0.355** [0.320, 0.391]. Controls behaved as expected, so we **withdrew all lens-based claims**.

**Independent verification by a co-author**

Z.-J. Li's separately built, pre-registered pipeline first reproduced the official lens on Qwen3.5-0.8B (gap 0.001 vs. 0.038 split-half). On Qwen3-1.7B direct harmful requests, adding Jacobian-lens features changed AUPRC by only −0.0015 to −0.0004, and lens-based scores trailed logit-lens ones (ΔAUROC −0.134 to −0.043).

### Finding 3 · The outcome is settled late




The signal grows with depth and peaks at layer 20; the refusal-direction projection becomes informative only at layer 20 (0.659 vs. 0.530 at layer 14). Best setting per layer, train+dev CV.

### Conclusion

1. **Lens behavior does not transfer between SLMs.** A fitting procedure that reproduced the official lens on Qwen3.5-0.8B added no measurable signal on Qwen3-1.7B.
2. **The two lenses differ in kind.** Where a guard reads, the simpler logit lens tracked the model better.
3. **Validate before you guard.** Use a lens validated for the model and position; until then, guards should read the raw hidden state.

Results are preliminary: provisional labels, train+dev only.

**References.** Jacobian lens: W. Gurnee et al., arXiv:2607.15495, 2026 · Qwen3: A. Yang et al., arXiv:2505.09388, 2025 · JailbreakBench: P. Chao et al., arXiv:2404.01318, 2024 · HarmBench: M. Mazeika et al., arXiv:2402.04249, 2024 · StrongREJECT: A. Souly et al., arXiv:2402.10260, 2024 · Refusal direction: A. Arditi et al., arXiv:2406.11717, 2024